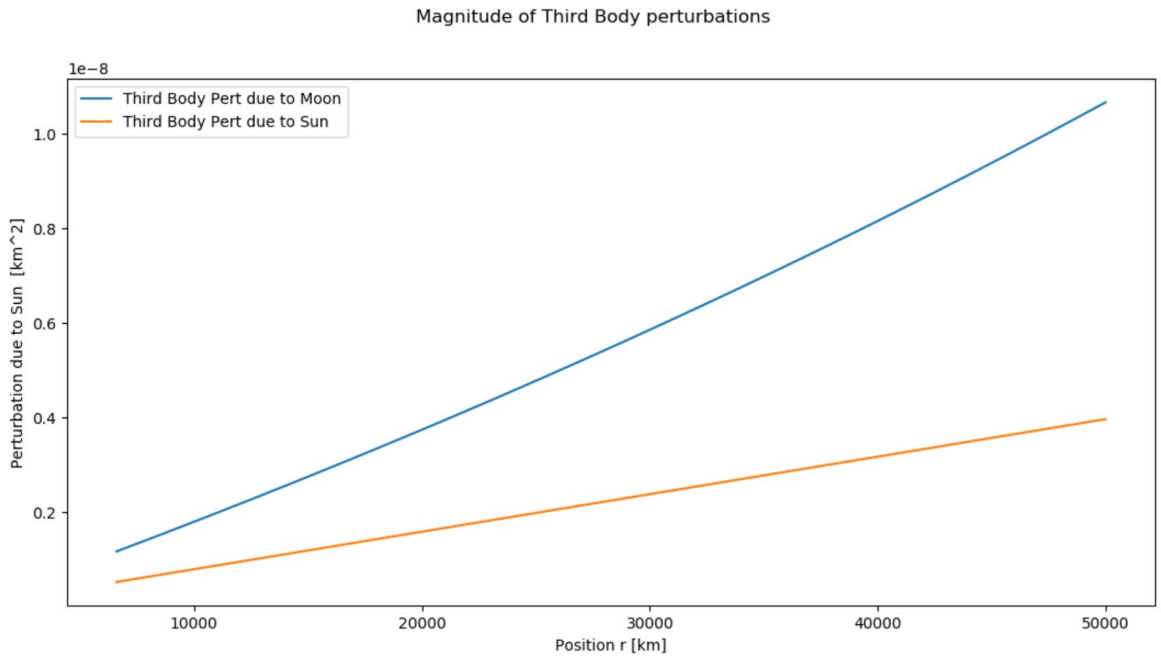
Problem 3

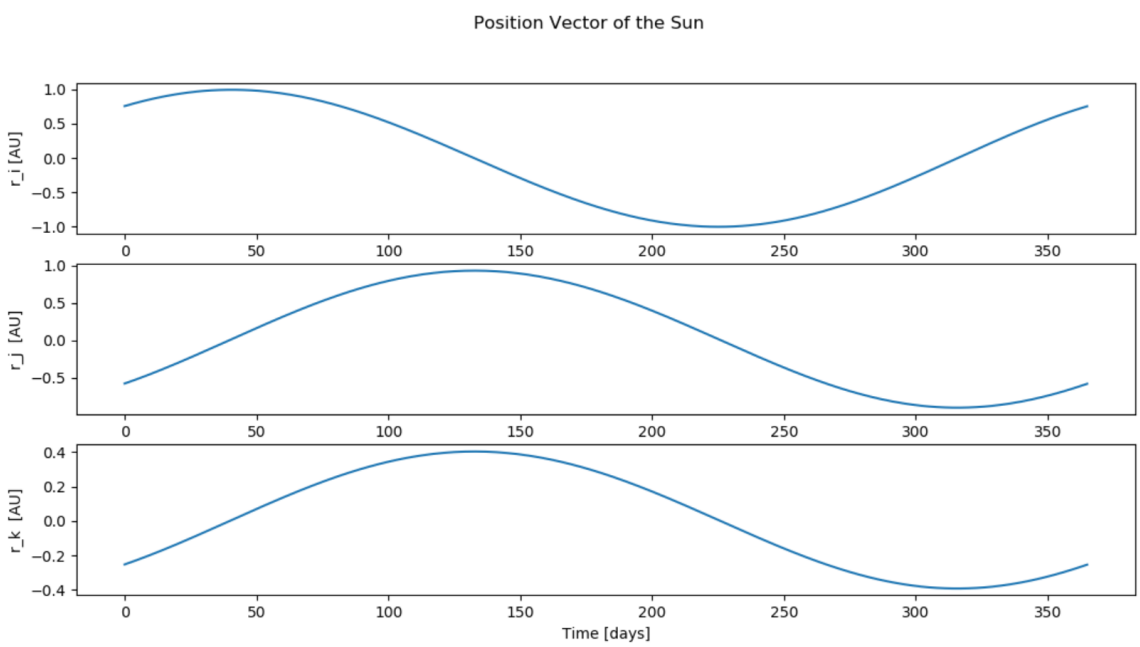


Problem 4

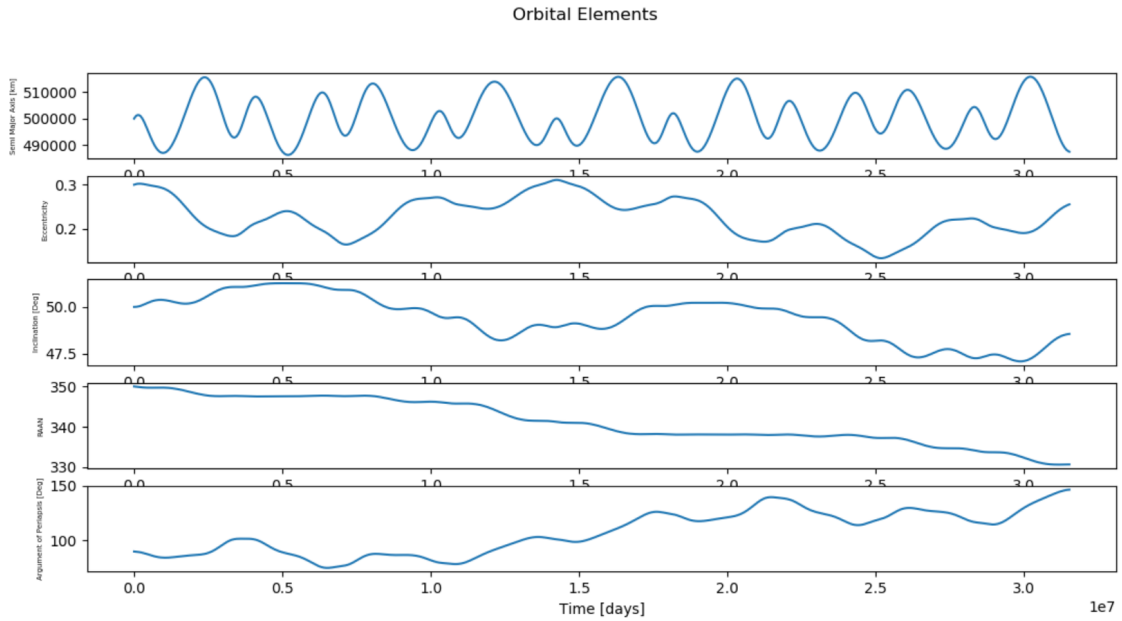
Problem 4a

Position in GCRF Frame [AU]: [[0.75780975 -0.57960912 -0.25126748]]

Problem 4b

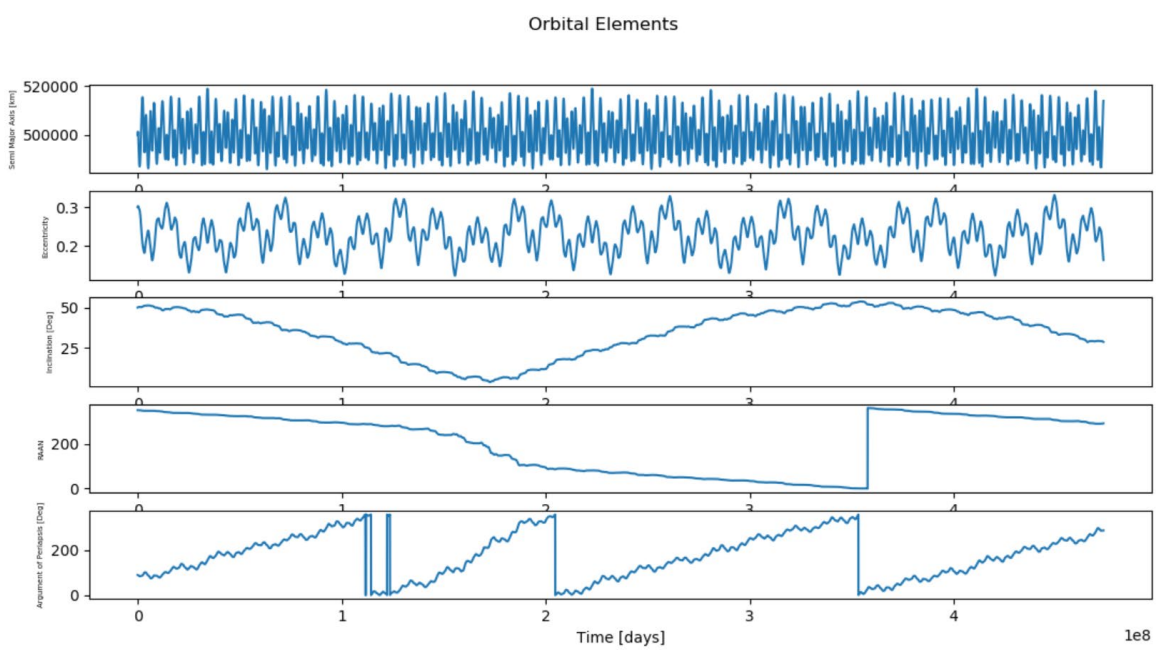


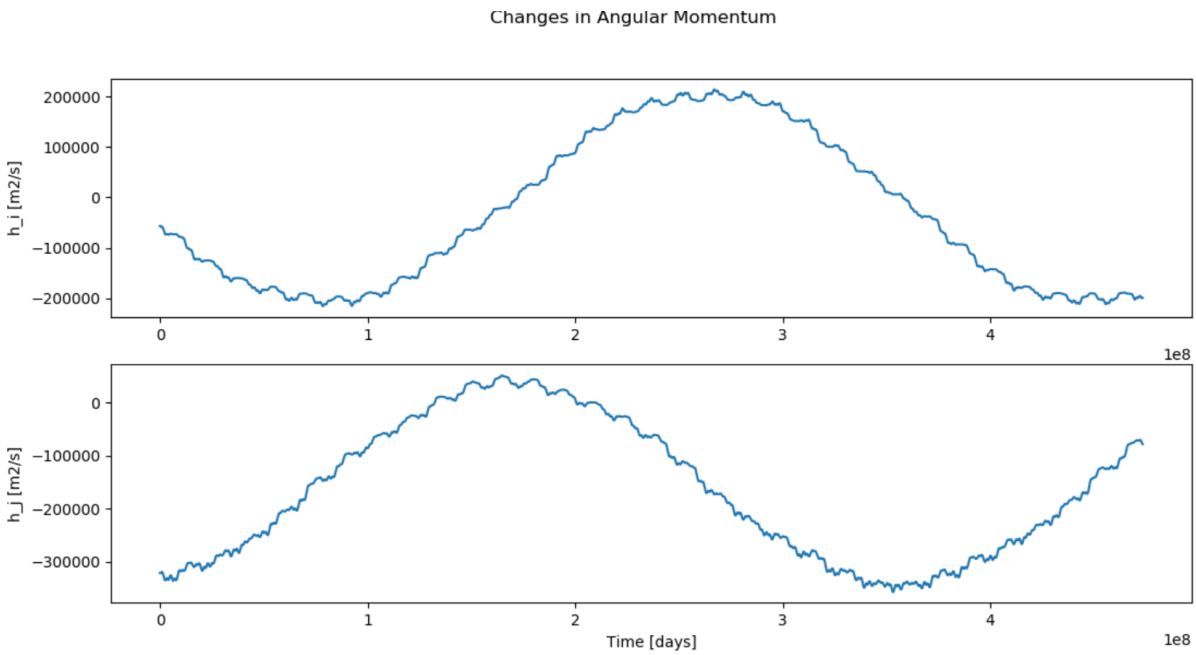
Problem 5



Yes! The changes in orbital elements match our expectations. Due to the perturbations of the sun, the orbit is changing in magnitude (as seen by the eccentricity and semi major axis). Based on our classroom discussions, the graphs above match our expectations that RAAN is changing. Due to the oblateness of the Earth, the force from the sun will exhibit a torque, thus causing the Earth's khat vector to experience a precession.

Bonus Question




Changes in Angular Momentum
h_i [m2/s]
200000
100000
0
−100000
−200000
0
1
2
3
4
1e8
h_j [m2/s]
0
−100000
−200000
−300000
0
1
2
3
4
Time [days]
1e8